# From AF scores to clinical decisions

Oxford CDT · Machine Learning for Time Series · Clinical Utility Practical

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| Date and time | Friday 6 November 2026, 14:30–16:00 |
| Organiser(s) | Xiao Gu |
| Instructor(s) | Bradley Segal |
| Demonstrator(s) | Xiao Gu, Edward Phillips |
| Course page | github.com/BradSegal/OxfordCDT2026ClinicalUtility |

## What you will do

A model gives each ECG recording a score and a service has to decide which recordings to send for AF review. You will price the four outcomes of that decision, derive a threshold from the prices, write the function that prices a set of decisions, compare two models at the same prices, recalibrate a score, and freeze a policy before reading its evaluation report. Recording labels come from PhysioNet/CinC 2017. Prices are in reviews, where a review costs one.

## Getting started

1. Open the course page and click the **Open in Colab** badge for the core practical.
2. Log in to Google (any account will work, including a personal one).
3. Click **File > Save a copy in Drive** and work within the notebook that opens.
4. Run the setup cell. If Colab says the notebook "was not authored by Google", click **Run anyway** to confirm that you trust the code. Wait for `Course helpers ready.`
5. You can run the cells sequentially by pressing **Shift+Enter**. A CPU runtime is enough.

## The notebooks

| Notebook | When | Time |
|---|---|---|
| Core: From AF scores to clinical decisions | In the session | The session |
| Extension 1: Calibration: Platt, isotonic and prior shift | Self-paced | About 35 min |
| Extension 2: Comparing models: reclassification and net-benefit differences | Self-paced | About 30 min |
| Extension 3: Class weighting versus moving the threshold | Self-paced, optional | About 30 min |
| Extension 4: Capacity, prevalence and daily queues | Self-paced | About 30 min |
| Extension 5: Window-level calls and unreadable recordings | Self-paced, optional | About 35 min |
| Extension 6: Your own classifier's scores | Self-paced | About 25 min |

The rough timing is to aim to be set up by 14:40, work through the core content independently, then participate in the closing discussion from about 15:50. Please note that these times are rough guides and there is no issue if you do not finish, as the notebook is yours to keep. Afterwards, you can proceed to Extension 1 if you had any issues with calibration or the decision curves, Extension 2 to compare models, Extension 4 for deployment, or Extension 6 to use your own classifier's scores. Worked copies are in the respective `solutions/` folders.

## How the problems work

Each problem is labelled either **Make** or **Fix the bug**. Replace the `...` with your answer and run the cell. If you are correct, the checker prints ✓. If the answer is incorrect, you will receive **Not yet** with a hint that names the likely mistake. If you see **Not attempted yet**, the `...` is still in the answer. Cells without a problem run library code for you; read the note after each one. Nothing is collected or marked. If a slider does not appear, you can use the ordinary function call in the next cell and modify its numbers. If you receive a `NameError`, please use **Runtime > Run before**.

If you get stuck for about 10 minutes, just move on. In the core, you can delete the # from the commented `check.use_reference(...)` line below Problems 1, 3, 5 and 6 to use the reference version; in the extensions, the same switch sits below each answer a later cell uses. A cell that prints **Waiting for** names the problem it needs: once this is completed or switched, click on the waiting cell, use **Runtime > Run before**, then run it. Worked copies are in the `solutions/` folders.

## The decision matrix

| | Reference AF | No reference AF |
|---|---|---|
| **Review** | True positive (TP): $C_{TP} = 1$ | False positive (FP): $C_{FP} = 1$ |
| **No review** | False negative (FN): $C_{FN} = 5$ | True negative (TN): $C_{TN} = 0$ |

TP/FP/FN/TN indicate the number of recordings for each outcome, and $C_{TP}$/$C_{FP}$/$C_{FN}$/$C_{TN}$ indicate the price of a single recording for that outcome, in reviews. As such, even a correct review still incurs a review, so $C_{TP}$ is not zero. The ordering returned by scikit-learn's `confusion_matrix(y, review, labels=[False, True]).ravel()` is `tn, fp, fn, tp`.

## Threshold probability

For a recording with AF probability p, the expected loss of reviewing it is identical to that of not reviewing it at the threshold probability

$$p_t = (C_{FP} - C_{TN}) / [\,(C_{FP} - C_{TN}) + (C_{FN} - C_{TP})\,]$$

The numerator indicates the harm of reviewing a recording without AF and the second bracket indicates the benefit of reviewing one with AF. Given the default prices, this means that $p_t = 0.2$. Increasing the price of a miss would result in a reduction in $p_t$. Note that a model's score is only a true probability if the model is calibrated.

## Net benefit

The net benefit of a model's predictions as compared to reviewing nobody, at threshold probability $p_t$, is defined as:

$$NB(p_t) = TP / n - (FP / n) \times p_t / (1 - p_t)$$

This metric produces a value in true-positive equivalents per recording. As an example, at $p_t = 0.2$ the relative weight on false positives is 0.25, which matches the priced matrix. Read a decision curve vertically by fixing the preference first and then comparing the different strategies at that value.

## Categorical NRI

Reclassification from an old model to a new one at a particular cutoff, counted within each outcome group:

$$NRI_{event} = P(up \mid AF) - P(down \mid AF)$$
$$NRI_{non\text{-}event} = P(down \mid no\ AF) - P(up \mid no\ AF)$$
$$NRI = NRI_{event} + NRI_{non\text{-}event}$$

This metric adds two proportions and treats them as equally valuable. It is independent of prices and so may result in a positive NRI in the face of a worse net benefit.

## Keep the denominators visible

- **Sensitivity** TP / (TP + FN): of all the recordings with AF, how many were reviewed?
- **Specificity** TN / (TN + FP): of all the recordings without AF, how many were not reviewed?
- **PPV** TP / (TP + FP): of all the recordings that were reviewed, how many had AF?
- **NPV** TN / (TN + FN): of all the recordings that were not reviewed, how many did not have AF?

A rate with an empty denominator is not zero, it is undefined. "No reference AF" also does not mean healthy, it includes other rhythms as well.

## Finish with a decision

Freezing your model, cutoff and prices is vital before you look at the evaluation split. While you can see how a cutoff chosen on development performs on development, these results are apparent; the evaluation report, with its intervals, shows how the frozen policy performs on recordings that played no part in choosing it. The final step is to write up the decision record. This should include the action, the prices you selected and why, the specific model and cutoff, both reports, as well as the change that would make you reopen the decision.

Please note, this is teaching code and must not be used for making decisions about real patients.